Machine Learning, Mid Sem, Answer any four questions out of six

1. (a) State the difference between supervised and unsupervised learning. (3)

   (b) If Gradient Descent is diverging rather than converging, what is one possible technique to resolve this issue? (2)

2. (a) What happens if we apply logistic regression to a dataset that is not linearly separable? (2)

   (b) How do you interpret the output probability of a logistic regression model? (3)

3. (a) Why is KNN called a lazy learning algorithm? (2)

   (b) How does the choice of 'K' affect the performance of the KNN algorithm? (3)

4. A company wants to classify whether a customer will purchase a product based on two features: Age Group and Income Level. For a given dataset, If a new customer is Middle-Aged with High Income, use the Naïve Bayes classifier to predict whether they will purchase the product.

Age_Group	**Income_Level**	**Purchase (Yes/No)**
Young	Low	No
Young	Medium	Yes
Young	High	No
Middle-Aged	Low	Yes
Middle-Aged	Medium	Yes
Middle-Aged	High	No
Senior	Low	No
Senior	Medium	Yes
Senior	High	Yes

5. A hospital is testing a new ML-based diagnostic tool on 100 patients. Of these, 50 have the disease, and 50 do not. The test correctly identifies 40 sick patients but misses 10. It also wrongly classifies 20 healthy patients as sick while correctly marking 30 as disease-free. Calculate the accuracy, precision, and recall of the ML-based diagnostic tool.

6. A dataset consists of 16 instances, evenly distributed between two classes: Class A and Class B, with 8 instances each. After splitting the dataset based on attribute X, the data is divided into two subsets. The left node contains 6 instances, with 4 belonging to Class A and 2 to Class B, while the right node consists of 10 instances, with 4 from Class A and 6 from Class B. Calculate the information gain resulting from this split by using the entropy.